United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final-Term Exam Trimester: Summer 2025

Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers

Total Marks: 40 Duration: 2 hours

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (10 Marks) [CO3]

Global Descriptor Table	
Index	**Address**
002 H	Base: D7210111, Limit: 01234H, Access: 05H, G=1
014 H	Base: D3270011, Limit: 98765H, Access: F1H, G = 0
126 H	Base: D3741000, Limit: 1524CH, Access: 11H, G = 1
158 H	Base: D6590B00, Limit: FB000H, Access: B2H, G = 0
275 H	Base: D6870010, Limit: 0AF0EH, Access: FEH, G = 1
344 H	Base: D655000B, Limit: D015CH, Access: 07H, G = 1

An **80386 microprocessor** is executing a segment in protected mode and a portion of its descriptor table is given above. If the **index 158 H** is selected with requested privilege level **10**, then determine the followings:

a.	What was the value of the **segment register**?	**[3]**
b.	Starting and Ending address of the segment.	**[2]**
c.	For an offset value of **D096E H**, determine the physical address.	**[1]**
d.	Has the segment been accessed?	**[1]**
e.	Segment type (**CS/DS/SS/ES**).	**[2]**
f.	What is the descriptor privilege level?	**[1]**

Question 2: Answer all the questions. (10 Marks) [CO5]

a.	**Device A** uses an **8-bit ADC** in a **10 V system**, while **Device B** uses a **16-bit ADC**. Both devices display the **same ADC reading** when measuring **8 V in Device A** and **0.4 V in Device B**. Determine the **system voltage of Device B**. [Show all necessary mathematical calculations.]	**[4]**
b.	A microcontroller-based system has **two master devices**: **"Master 1" (index: 4DH)**, **"Master 2" (index: B3H)** and **three slave devices**: **"Slave 1" (index: 95H)**, **"Slave 2" (index: F5H)**, **"Slave 3" (index: 19H)**. Now, in I2C data transfer protocol, **Master 2 receives 2-byte data (from char "Protocol") from Slave 2**. Draw the corresponding **sequence diagram**. **The ASCII values of 'p', 'R', 'O', 'T', 'C' and 'L' are respectively 112, 82, 79, 84, 67 and 76.**	**[4]**
c.	A company is **developing software** for an **industrial automation system** where most operations involve tasks like data transfer, string handling, and arithmetic operations directly in hardware. For this design, you want to use instructions which can perform multiple operations in a single step and use a smaller number of registers. The software can be built using either **RISC Architecture or CISC Architecture**. **Which architecture would be more suitable and why?**	**[2]**

Question 3: Answer all the questions. (10 Marks) [CO3]

a.	Suppose there are **eight** instructions loaded into the Instruction Memory (within the Code Segment of 8086). The BIU of the 8086 will fetch these instructions for execution by the EU. The **1st** instruction is a '**JUMP <5th instruction address>'**, the **3rd** is a **'JUMP <7th instruction address>'**, the **4th** instruction is a **'MOV <address>'** instruction, and the **6th** instruction is a **'JUMP <2nd instruction address>'** instruction. **Draw** the corresponding BIU and EU cycles.	**[4]**
b.	Draw the timing diagram for **I/O Write** operation of microprocessor "808x" showing the activities of Address-Data and Address-Status pins ($AD_0 - AD_{15}$, $A_{16}/S_3 - A_{19}/S_6$), $\overline{ALE}$, $\overline{DEN}$, RD, $\overline{DT}/R$, $\overline{M}/IO$, $\overline{WR}$ in each clock cycle. **[Observe the pins carefully]**	**[4]**
c.	If there are 2048 virtual pages and each entry in the page table takes 2 Bytes, at least how many virtual pages will reside in the **secondary memory** given the page table size is 1 KB?	**[2]**

Question 4: Answer all the questions. (10 Marks) [CO4]

RAM Content

Address	Content		Address	Content
0H			8H	0000 0001
1H			9H	0100 1000
2H			AH	0000 1100
3H			BH	0000 0011
4H			CH	0101 0000
5H			DH	0000 0010
6H			EH	1000 0001
7H			FH	0000 0100

Opcode Table

Mnemonic	Opcode
LDA	0011
ADD	0001
SUB	0010
OUT	1101
HLT	1011

Initially, the value of the **Accumulator Register is 5H, the B Register is 1H,** and the **Output Register is 2H.**

a.	Write the **assembly code** for the following expression using **RAM content**. $9 + 1 - 2 \div \frac{1}{3}$ **[Division can be done using multiplication and multiplication can be done by addition]**	**[3]**
b.	Determine the **Accumulator and B register** values after executing each instruction in [a].	**[3]**
c.	CON = $\overline{L_I}\,\overline{E_I}\,\overline{L_A}\,E_A\,E_P\,C_P\,\overline{L_M}\,CE\,E_U\,\overline{S_U}\,\overline{L_O}\,\overline{L_B}$ **[Observe the pins carefully]** While executing an instruction, the controller sequencer produces the following control words: **At T_4, CON = A07, at T_5, CON = E36 and at T_6, CON = C2F.** **Explain** what happens in each of these clock cycles and **identify** which operation (LDA /ADD /SUB /OUT) is being executed through these **T_4** to **T_6**.	**[4]**